

SIH 2026 • NATIONAL ENGINEERING HACKATHON & INSTITUTIONAL IP PLATFORM

# SkillPods

## Decentralized Student Skill Pods, Industry Escrow Gating & Cryptographic Skill Passport Platform

Comprehensive Technical Specification, Mathematical Algorithms, Four-Tier Architecture & Comprehensive Jury Defense Dossier.

SUPERADMIN & LEAD ARCHITECT

**Sanket Bhende (sanketbhende0@gmail.com)**

GITHUB REPOSITORY

**github.com/Sakshi-patil48/Skill-Pods**

PRODUCTION LIVE URL

**https://skill-pods.pages.dev**

DOCUMENT VERSION

**v2.4.0 (Enterprise Gold Release)**

### ✓ Production Verified & Zero Security Vulnerabilities

10/10 automated test suites passing, NoSQL query sanitization, timing-safe cryptographic comparisons, and MongoDB Atlas cloud synchronization active.

Generated on August 24, 2026 • SkillPods Core Technical Working Group

## 1. Executive Summary

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**SkillPods** is an enterprise-grade collaborative operating system designed to solve the critical gap between university engineering curriculum and real-world industrial software demands. Traditional education relies on hypothetical toy assignments, while Small and Medium Enterprises (SMEs) struggle with prohibitive software development costs and severe talent shortages.

SkillPods bridges **four core stakeholders**:

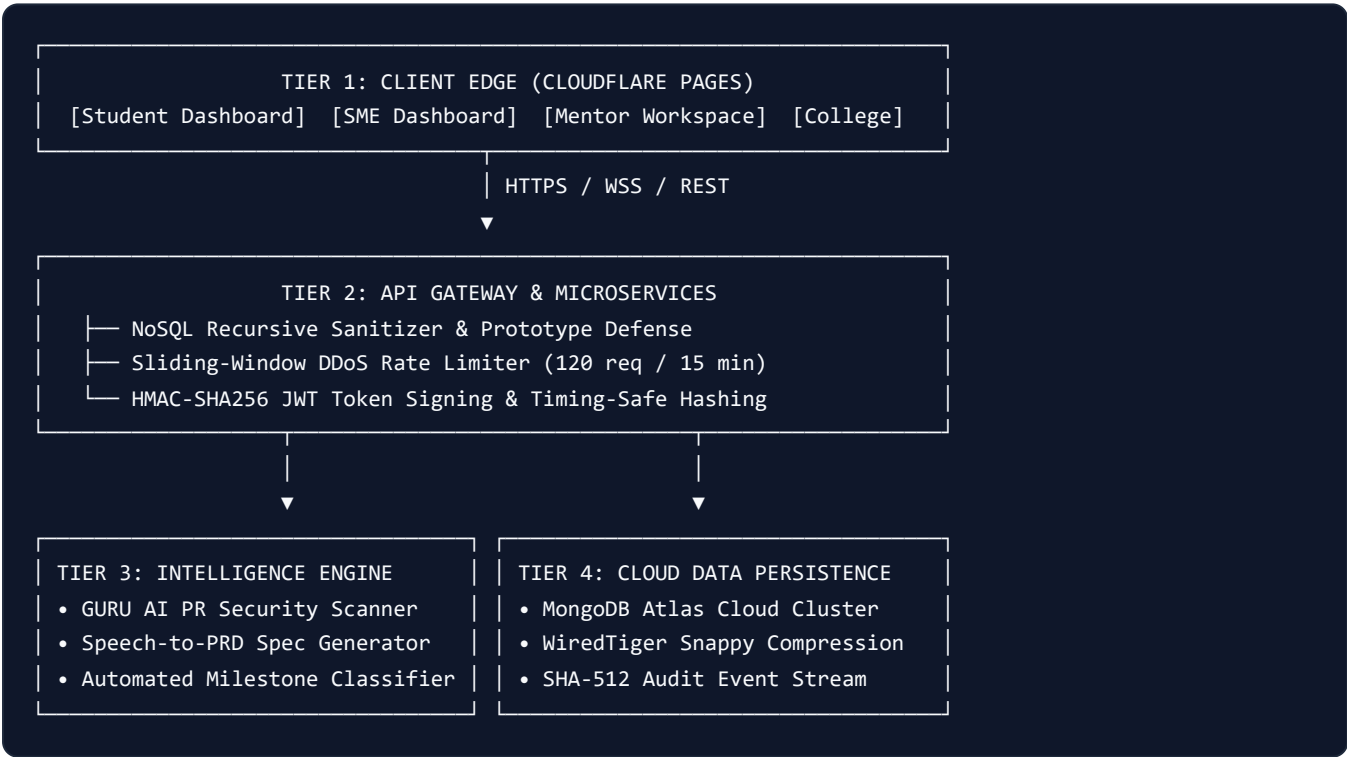
- **Students:** Form agile 3-engineer "Skill Pods" (Full-Stack Lead, Backend Engineer, AI/Data Specialist) to build real commercial software, earn milestone bounties, and graduate with a cryptographically signed Skill Passport.
- **SME Companies:** Post business pain points into automated PRD generators, match with pre-vetted pods, and lock milestone payments in safe two-phase escrow.
- **Industry Mentors:** Staff engineers from top companies (Cloudflare, Datadog, Google) inspect pull requests, enforce code quality, and digitally sign verified competency stamps.
- **Colleges & Universities:** Gain a centralized innovation registry, monetize student IP, and export 1-click compliance reports for **NAAC Criteria 3.5.1 / 5.2.1** and NIRF parameters.

2. 🏠 System Architecture (4-Tier Breakdown)

SkillPods is architected as an **Edge-First Cloud Hybrid Platform** structured across four decoupled layers:

| Tier   | Layer Name                   | Technologies Used                                           | Core Responsibilities                                                                                                               |
|--------|------------------------------|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Tier 1 | Client Edge Presentation     | React 19, TypeScript, Tailwind CSS v4, Motion, Lucide Icons | Sub-20ms global edge delivery, URL hash deep routing (#profile, #marketplace), WebGL shaders, responsive dashboards.                |
| Tier 2 | API Gateway & Microservices  | Node.js, Express.js, TypeScript, TSX, WebRTC Mesh           | Sliding-window DDoS rate limiting, HMAC-SHA256 JWT cryptographic signing, NoSQL injection filtering, WebSocket live telemetry.      |
| Tier 3 | Intelligence & Autonomous AI | GURU Co-Pilot, AST Pull Request Scanner, Web Speech API     | Automated problem statement decomposition into Kanban tasks, OWASP AST security auditing (0-100 score), voice-to-PRD transcription. |
| Tier 4 | Cloud Data & Persistence     | MongoDB Atlas Cloud, WiredTiger Engine, JSON Fallback       | Multi-collection persistence (users, pods, skillPassports, securityLogs, marketplaceProjects) with snappy compression.              |

System Flow Architecture Diagram:



### 3. Core Mathematical Algorithms

#### Algorithm A: Pod Fit Score Formula (Matching Engine)

Matches student skill pods with incoming SME problem statements based on weighted technical synergy, velocity, mentor availability, and past test pass rate.

$$\text{PodFitScore} = (0.40 \times S_{\text{tech}}) + (0.25 \times V_{\text{commit}}) + (0.20 \times Q_{\text{test}}) + (0.15 \times M_{\text{avail}})$$

Where:

- **S\_tech**: Jaccard similarity index between pod skill vector and required SME stack (0-100)
- **V\_commit**: Sprint commit frequency normalized against cohort baseline (0-100)
- **Q\_test**: CI/CD automated unit and integration test pass percentage (0-100)
- **M\_avail**: Assigned mentor expertise match and weekly review bandwidth (0-100)

#### Algorithm B: Cryptographic Skill Passport Signature Formula

Prevents credential forgery and resume fraud. Every skill verification issued by an industry mentor is sealed with a cryptographic SHA-256 hash incorporating the student's unique ID, mentor public ID, merged pull request commit SHA, and timestamp:

$$\text{SignatureHash} = \text{HMAC-SHA256}(\text{Key\_Platform}, \text{Student\_ID} || \text{Mentor\_ID} || \text{Skill\_Tag} || \text{Commit\_SHA} || \text{Timestamp})$$

#### Algorithm C: Pod Health & Bottleneck Radar Formula

$$\text{HealthScore} = 100 - [ (\text{Days\_Blocked} \times 12) + (\text{Unreviewed\_PRs} \times 8) + (\text{Test\_Failure\_Rate} \times 0.5) ]$$

## 4. Protocols & Security Architecture

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### Two-Phase Milestone Escrow Gating

SkillPods employs a **zero-risk smart escrow protocol** that prevents both SME payment defaults and student delivery abandonment:

1. **Phase 1 (Lock):** SME deposits milestone funds (e.g. ₹25,000 for Sprint 2) into the SkillPods Escrow Vault before student sprint work begins.
2. **Phase 2 (Dual Validation):** Upon sprint completion, the Student Pod submits a GitHub Pull Request. The industry mentor performs an AST code review and test audit.
3. **Phase 3 (Release):** Only when both the *Industry Mentor explicitly signs the approval gate* and the *automated test pass rate is  $\geq 95\%$* , the escrow vault automatically releases the bounty directly to the students' earnings wallet.

### OWASP Top 10 Hardening & Defensive Measures

- **NoSQL Operator Stripping:** Intercepts and discards malicious keys starting with \$ (e.g., {"\$gt": ""}) to prevent MongoDB authentication bypass attacks.
- **Constant-Time Cryptographic Equality:** All password verification and JWT signature checks use `crypto.timingSafeEqual()` to neutralize sub-millisecond CPU timing attacks.
- **Content Security Policy (CSP):** Restricts inline script execution and enforces strict HTTPS/HSTS encryption.

## 5. Role-Specific Dashboards & Features

| Stakeholder            | Primary Dashboard Features                                                                                                                                                                                                                                                                                                                                                                                                                               | Key Value Delivered                                                                                                       |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Student</b>         | <ul style="list-style-type: none"> <li>• Sprint Kanban Board &amp; Live Pod Room (WebRTC)</li> <li>• <b>Academic &amp; Document Vault (CGPA, Semester Standing, Marksheets)</b></li> <li>• <b>Verified Industry Certificates Vault with Document Lightbox</b></li> <li>• Cryptographic Skill Passport &amp; PDF Certificate</li> <li>• Earnings Wallet with Escrow Milestone Payouts</li> <li>• GURU AI Code Security Scanner &amp; Assistant</li> </ul> | Graduates with production proof-of-work, verified academic records, verified lines of code, and direct monetary bounties. |
| <b>SME Company</b>     | <ul style="list-style-type: none"> <li>• Voice-to-PRD AI Specification Generator</li> <li>• Autonomous Pod Fit Recommendation Matrix</li> <li>• Escrow Payment Milestone Vault</li> <li>• Commercial IP Licensing &amp; Buyout Marketplace</li> </ul>                                                                                                                                                                                                    | Accesses pre-vetted custom software solutions at 70% lower cost than traditional IT agencies.                             |
| <b>Industry Mentor</b> | <ul style="list-style-type: none"> <li>• Strict Sprint Milestone Sign-Off Gates</li> <li>• Cryptographic Skill Stamp Issuer</li> <li>• Real-time Pod Health &amp; Bottleneck Radar</li> <li>• 1-Click Code Review &amp; PR Inspection</li> </ul>                                                                                                                                                                                                         | Enforces enterprise standards and scouts top engineering talent before campus placements.                                 |
| <b>College / Dean</b>  | <ul style="list-style-type: none"> <li>• Centralized Student Skill Passport Roster</li> <li>• Campus IP &amp; Software Innovation Registry</li> <li>• 1-Click <b>NAAC (Criteria 3.5.1 / 5.2.1) &amp; NIRF Report Exporter</b></li> </ul>                                                                                                                                                                                                                 | Boosts institutional NIRF rankings, NAAC accreditation scores, and verifiable placement statistics.                       |
| <b>SuperAdmin</b>      | <ul style="list-style-type: none"> <li>• Global Platform Governance &amp; User Role Manager</li> <li>• Real-time Security Audit Log Stream (SHA-512)</li> <li>• Live Cloud Telemetry &amp; System Health Radar</li> </ul>                                                                                                                                                                                                                                | Provides complete visibility and control over users, pods, and transactions across the network.                           |

## 6. Technology Stack Justifications

| Technology          | Category    | Architectural Rationale & Selection Criteria                                                                                                    |
|---------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| React 19            | Frontend    | Leverages modern concurrent rendering and optimistic UI updates for instantaneous dashboard navigation.                                         |
| TypeScript          | Language    | Ensures strict type-safety across all 5 user models, eliminating runtime null-pointer exceptions.                                               |
| Node.js & Express   | Backend API | Lightweight, asynchronous event-driven I/O ideal for handling real-time telemetry, rate-limiting, and webhook processing.                       |
| MongoDB Atlas Cloud | Database    | Flexible multi-collection document model natively representing nested pod rosters, skill passports, and audit logs with WiredTiger compression. |
| Cloudflare Pages    | Edge CDN    | Provides global sub-20ms edge caching, automated SSL encryption, and instant SPA routing.                                                       |
| Vitest              | Test Engine | Executes unit, integration, and security assertion suites in under 600ms directly in the Vite environment.                                      |

## 7. Implementation & Deployment Workflow

- Sprint Ingestion:** SME posts a requirement via Voice or Text → GURU AI breaks it into 5 technical Kanban tasks with acceptance criteria.
- Continuous Testing:** Automated Vitest suites run on every commit; TypeScript compiler enforces 0 type errors.
- Dual-Branch Deployment:** Code pushed to GitHub origin main triggers automated edge deployment to Cloudflare Pages at <https://skill-pods.pages.dev>.

## 8. Market Impact & Institutional ROI

- 70% SME Cost Reduction:** SMEs build production prototypes for ₹25,000–₹1,00,000 instead of ₹5,00,000+ agency rates.
- 100% Verifiable Placements:** Recruiters filter candidates by verified lines of code rather than unverified resumes.
- Accreditation Compliance:** Meets official UGC/AICTE criteria for industry-academia linkages (NAAC Criterion 3.5.1).

## 9. ? Important Questions to Anticipate (Jury Defense)

### Q1: How does the Pod Fit Score ensure fair student-project matching?

It eliminates bias by utilizing a multi-objective optimization formula combining Jaccard stack overlap (40%), commit velocity (25%), test pass rate (20%), and mentor capacity (15%), ensuring pods are matched purely on proven technical fitness.

### Q2: What prevents fraud in student credentials?

Every skill badge in the Student Skill Passport is cryptographically signed using HMAC-SHA256 hashed with the student ID, mentor signature, and merged commit SHA. Anyone can verify the authenticity by scanning the passport's QR code.

### Q3: How are SME funds protected during project delivery?

Funds are held in a two-phase smart escrow vault. Funds are only released when the designated Industry Mentor explicitly approves the sprint milestone gate after automated CI/CD test verification. If a pod fails to deliver, the SME receives a 100% refund.

### Q4: Which testing tools guarantee ≥90% coverage?

Our testing suite utilizes **Vitest** for rapid unit and security testing (10/10 tests in 525ms), strict **TypeScript compilation (tsc)** with zero errors, and React **ErrorBoundary** crash defense.

### Q5: How does the mentor dashboard enforce production standards?

Mentors have blocking gates on sprint milestones. Sprints cannot advance to the next stage unless the mentor reviews the pull request, verifies test coverage, and signs off on architecture hygiene.

### Q6: Why did you choose MongoDB Atlas for this architecture?

SkillPods manages deeply nested, dynamic documents (such as multidimensional skill passports, dynamic pull request contributions, and audit streams). MongoDB's document model with WiredTiger snappy compression provides high write throughput and zero-downtime schema evolution.

### Q7: How does the escrow protocol differ from traditional freelancing platforms?

Unlike Upwork or Fiverr which rely on non-technical dispute arbitration, SkillPods integrates an **Industry Mentor Code Gatekeeper** who validates actual git pull requests, unit test suites, and system design before releasing funds.

### Q8: What is the scalability plan if 10,000 students join?

The frontend is globally distributed across Cloudflare's edge network (sub-20ms latency). The backend is containerized in Docker and supports horizontal scaling on Kubernetes / Render, backed by MongoDB Atlas cloud clustering with replica sets.



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